

Vanishing Twins

Time Limit: 1.0 seconds

Memory Limit: 256 megabytes

Alice and Bob are playing a game on a dynamic multiset of numbers. You are given an array A of N integers.

For each prefix of length k ($1 \leq k \leq N$), a standalone game is played on a board initially containing the multiset of elements $\{A_1, A_2, \dots, A_k\}$.

The game proceeds in turns with Alice going first:

1. On a player's turn, they must choose two distinct indices i and j currently on the board such that $A_i = A_j$.
2. The player then removes the element at index j from the board.
3. The game ends when no two remaining elements on the board are equal (i.e., all elements on the board are pairwise distinct).

Alice plays to maximize the sum of the remaining elements on the board at the end of the game, while Bob plays to minimize this sum.

Assuming both players play with optimal strategy, determine the final sum of the elements remaining on the board for every prefix length k from 1 to N .

Input Format

The first line contains an integer T ($1 \leq T \leq 10^4$) — the number of test cases.

For each test case:

The first line contains an integer N ($1 \leq N \leq 2 \times 10^5$) — the number of elements.

The second line contains N space-separated integers $A_1, A_2, \dots, A_N$ ($1 \leq A_i \leq 10^9$).

It is guaranteed that the sum of N over all test cases does not exceed 2×10^5 .

Output Format

For each test case, print a single line containing N space-separated integers, where the k -th integer represents the final score of the board for the prefix $A_1, A_2, \dots, A_k$.

Sample 1

Input:

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2
5
1 2 2 3 1
4
5 5 5 5
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Output:

1 3 3 6 6
5 5 5 5

Note

In the first test case for prefix $k = 5$ ([1, 2, 2, 3, 1]):

The initial board contains {1, 2, 2, 3, 1}.

Turn 1 (Alice): Chooses $A_1 = 1$ and $A_5 = 1$, removing A_5 . Board becomes {1, 2, 2, 3}.

Turn 2 (Bob): Chooses $A_2 = 2$ and $A_3 = 2$, removing A_3 . Board becomes {1, 2, 3}.

No further valid moves exist since all elements are distinct. The remaining sum is $1 + 2 + 3 = 6$.